

Final Progress Report

Project Title: *Maze Navigator Adversarial AI*

Course: Foundations of Artificial Intelligence

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At this stage, we have completed the core architecture of our project and implemented the major AI components required to support both demonstration and evaluation. The project centers on an adversarial maze-navigation environment in which an agent must reach a goal while avoiding one or more pursuers. The environment now supports both static and dynamic mazes through a base Maze class and an extended DynamicMaze that periodically shifts walls while preserving reachability between start and goal. To support adversarial reasoning, we also implemented a formal GameState abstraction that represents turn order, terminal conditions, legal actions, utility, and heuristic evaluation, which is used by the adversarial search algorithms. On top of this framework, we have integrated multiple search and planning approaches, including Minimax, Alpha-Beta pruning, Expectimax, LRTA*, Hill Climbing, and Beam Search for the fully observable setting, as well as multiple pursuer behaviors Random, Greedy, Beam Search, and A* to create opponents of varying sophistication. These systems are all present in the codebase and currently runnable in the main simulation pipeline.

A substantial portion of our progress has also been devoted to extending the project beyond a standard grid-search exercise into a richer AI setting that captures uncertainty and environmental change. We implemented a partially observable environment through PartialMaze, where the agent no longer has direct access to the true maze and instead maintains a belief map updated through local perception within a configurable visibility radius. To operate under this uncertainty, we implemented three online agents in `online_agent.py`: an Online A* replanning agent, an Online LRTA* agent, and an exploration-based frontier agent that first maps the environment and then transitions to goal-directed navigation. In parallel, we enriched the environment with additional gameplay and decision-making complexity by introducing terrain and object types such as traps, power-ups, mud, water, and road cells, along with corresponding status effects on both the agent and pursuers. We also built a visual simulator in `visual_game.py` that supports fog of war, split-view belief visualization, dynamic walls, algorithm switching, and multiple pursuers, allowing the project to function not only as a codebase for experimentation but also as a live interactive demonstration of several core AI concepts. Taken together, these additions mean that the project now addresses adversarial search, online search, partial observability, and dynamic environments within a single integrated framework.

In addition to the environment and algorithms themselves, we have completed a significant evaluation pipeline through `experiments.py`. This module supports batch execution of both full observability and partial observability experiments over multiple maze sizes, obstacle densities, visibility radii, dynamic/static settings, repeated seeds, and pursuer strategies. The runner exports

raw trial-level results as well as several aggregated summaries, including grouped summaries by configuration, summaries by agent, summaries by agent-pursuer pair, and summaries stratified by visibility radius in the partial observability setting. It also generates figures such as bar plots, line plots, and heatmaps to support analysis of win rates, runtime, nodes expanded, and exploration coverage. The CSV files produced from our most recent experiment runs confirm that this pipeline is operational and already generating meaningful data products for the final report. This experimental component is one of the most important aspects of our progress, because it moves the project beyond implementation and toward a comparative empirical study of how different AI methods behave under pursuit, uncertainty, and dynamic environmental change.

Before the final submission, our remaining work is focused on refinement, robustness, and analysis rather than major new implementation. First, we plan to polish the visual simulator further, particularly with respect to interface clarity and the handling of edge cases in which agents or pursuers become trapped in oscillatory loops. Second, we will run a final, more systematic set of experiments and use the resulting CSV summaries and visualizations to support a stronger comparative discussion in the final write up. We want to emphasize how performance varies with stronger pursuers, reduced visibility, and dynamic wall movement, as these settings best showcase the distinct strengths and limitations of adversarial, online, and approximate search methods. Finally, we will continue improving repository level presentation by tightening documentation and ensuring that the code submission is clean, runnable, and easy to navigate. Overall, the project is no longer in an exploratory setup stage; the principal systems have been implemented and integrated, and the work remaining is aimed at final validation, presentation quality, and deeper analysis.